

STA 640 — Causal Inference

Chapter 3.1. Observational Studies

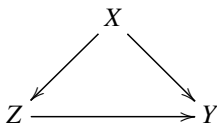
- Outcome Regression

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Observational Studies

- ▶ In observational studies, we do not control or know the assignment mechanism
- ▶ Confounding X is the main complication/hurdle between association and causation in observational studies



- ▶ Presence of measured and unmeasured confounders: usually unbalanced between groups, in contrast of RCT
- ▶ Measured confounders: also called covariates or pre-treatment variables or independent variables

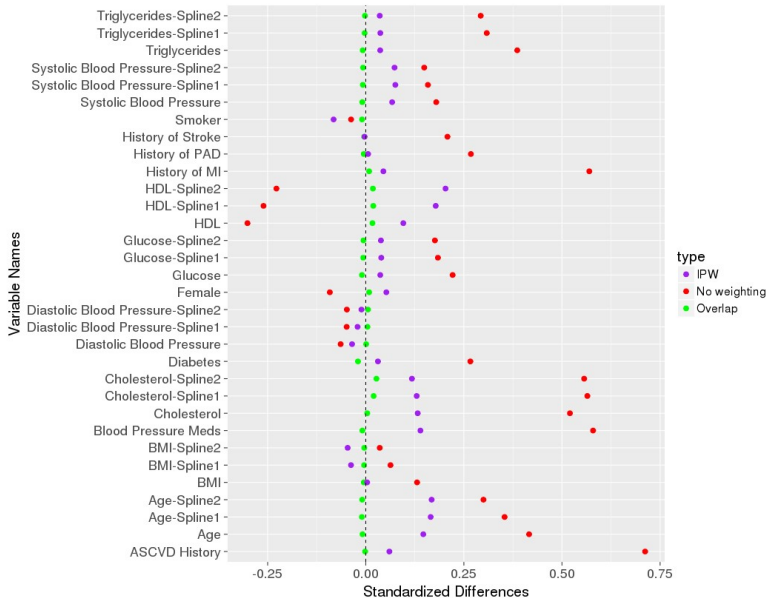
Example: Framingham Heart Study

- ▶ One of the longest-running observational studies
- ▶ Three cohorts first recruited in the early 1950s, 1970s and 2000s
- ▶ Total of over 15,000 participants with 5-60 years of follow-up
- ▶ Examined every 2-8 years
- ▶ Pioneer of cardiovascular risk prediction

Example: Framingham Heart Study

- ▶ **Patients:** cross-sectional population from the offspring cohort with a visit 6 (1995-1998)
- ▶ **Treatment:** statin use at visit 6 vs. no statin use
- ▶ **Outcomes:** CV death, myocardial infarction (MI), stroke Follow-up for outcomes begins on the date of visit 6, and is available for approximately 10 years
- ▶ **Confounders:** sex, age, body mass index, diabetes, history of MI, history of PAD, history of stroke, history of atherosclerotic cardiovascular disease, current smoking, systolic and diastolic blood pressure, blood pressure medications, total cholesterol, HDL cholesterol, triglycerides and fasting glucose

Visualize Covariate Balance in Framingham Study



Estimands

- ▶ **Conditional Average treatment effect (CATE):**

$$\tau(x) = \mathbb{E}[Y_i(1) - Y_i(0)|X_i = x].$$

- ▶ **Average treatment effect (ATE):**

$$\tau^{\text{ATE}} = \mathbb{E}[Y_i(1) - Y_i(0)].$$

- ▶ **Average treatment effect for the treated (ATT) or for the control (ATC):**

$$\tau^{\text{ATT}} = \mathbb{E}[Y_i(1) - Y_i(0)|Z_i = 1], \quad \tau^{\text{ATC}} = \mathbb{E}[Y_i(1) - Y_i(0)|Z_i = 0].$$

- ▶ For binary outcomes, **causal odds ratio (OR)**:

$$\tau^{\text{OR}} = \frac{\Pr(Y_i(1) = 1)/\Pr(Y_i(1) = 0)}{\Pr(Y_i(0) = 1)/\Pr(Y_i(0) = 0)}.$$

- ▶ For survival outcomes, **causal hazard ratio (HR)** (Cox proportional hazards model).

Estimands

- ▶ The relation between ATE, ATT, ATC:

$$\tau^{\text{ATE}} = \Pr(Z_i = 1)\tau^{\text{ATT}} + \Pr(Z_i = 0)\tau^{\text{ATC}}$$

- ▶ In randomized experiments, ATE is equivalent to ATT (and ATC) because treatment and control groups are comparable in expectation
- ▶ In observational studies, ATE is usually different from ATT and ATC
- ▶ The above relation does not hold for ratio estimands
- ▶ To identify the causal estimands, additional assumptions are required.

Identification Assumption 1: Unconfoundedness

Assumption 1: **Unconfoundedness (or ignorability)**

$$\{Y_i(0), Y_i(1)\} \perp Z_i | X_i$$

Equivalently, $\Pr(Z_i | Y_i(0), Y_i(1), X_i) = \Pr(Z_i | X_i)$

- ▶ Assumes that within subpopulations defined by values of observed covariates, the treatment assignment is random
- ▶ Rules out unobserved confounders, also known as the assumption of “no unmeasured confounders”
- ▶ Randomized experiments satisfy unconfoundedness
- ▶ Untestable in most observational studies, but sometimes can be indirectly tested, and sensitivity can be checked

Identification Assumption 2: Overlap

Assumption 2a: **Overlap (or positivity)**: for all i

$$0 < \Pr(Z_i = 1|X_i) < 1$$

Assumption 2b: **Strict overlap** : for some constant $\eta \in (0, 0.5)$, for all i

$$\eta < \Pr(Z_i = 1|X_i) < 1 - \eta$$

- ▶ $e(x) = \Pr(Z_i = 1|X_i = x)$: propensity score (Rosenbaum and Rubin, 1983)
- ▶ Overlap implies that, for all possible values of the covariates there are both treated and control units
- ▶ Important both conceptually and operationally (variance inflation)
- ▶ Overlap can be directly checked from the data
- ▶ Unconfoundedness and overlap jointly define the “**strong ignorability**” assumption (Rosenbaum and Rubin, 1983)

Overlap: Not as simple as it looks

- ▶ Overlap is as important as unconfoundedness in observational studies: **both** are required for identifying and estimating the causal effects
- ▶ In practice, the emphasis is usually on unconfoundedness and overlap is often taken for granted or ignored:
 - ▶ A notable example of ignoring overlap: “multiple causes” literature (Wang and Blei, 2019)
 - ▶ Claim to bypass unconfoundedness when there are multiple causes (treatments), but implicitly (and unknowingly) violates overlap
- ▶ Practically, with little overlap in covariates between treatments, reliable causal inference is practically impossible: extrapolation, large bias and variance, sensitive (more details in weighting)
- ▶ Overlap in observational studies with high-dimensional covariates: curse-of-dimensionality (D’Amour et al., 2021, JOE):
 - ▶ More covariates: (i) unconfoundedness is more plausible, but (ii) overlap is more difficult to satisfy

Nonparametric identification under strong ignorability

- ▶ Under strong ignorability, we have

$$\Pr(Y(z)|X) = \Pr(Y|X, Z = z)$$

The observed distribution of Y in treatment arm $Z = z$ equals the distribution of the potential outcome $Y(z)$ conditional on X .

- ▶ If treatment and control groups have the same X : as if randomized
- ▶ Under strong ignorability, we have two identification formulæ for ATE:

$$\begin{aligned}\tau^{\text{ATE}} &= E\{\mu_1(X) - \mu_0(X)\} \\ &= E\left\{\frac{ZY}{e(X)} - \frac{(1-Z)Y}{1-e(X)}\right\}\end{aligned}$$

where $\mu_z(X) = E(Y(z)|X) = E(Y|Z = z, X)$ is the **outcome model** under treatment $z(z = 0, 1)$.

Two/Three Estimation Strategies

- ▶ Two corresponding estimation strategies for ATE:

1. Outcome modeling (or regression) (Rubin, 1979):

$$\hat{\tau}^{reg} = N^{-1} \sum_{i=1}^N \{\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)\}$$

2. Inverse probability weighting (IPW) (Rosenbaum, 1987):

$$\hat{\tau}^{ipw} = \frac{\sum_{i=1}^N Z_i Y_i / \hat{e}(X_i)}{\sum_{i=1}^N Z_i / \hat{e}(X_i)} - \frac{\sum_{i=1}^N (1 - Z_i) Y_i / \{1 - \hat{e}(X_i)\}}{\sum_{i=1}^N (1 - Z_i) / \{1 - \hat{e}(X_i)\}},$$

where $\hat{e}$ is the estimated PS, $\hat{\mu}_z$ is the estimated outcome with X_i

- ▶ A third estimator is to combine outcome modeling and propensity score (IPW): augment one by another, known as the doubly-robust (DR) estimator (more in Chapter 3.5)

$$\hat{\tau}^{dr} = \hat{\tau}^{reg} + N^{-1} \sum_{i=1}^N \left\{ \frac{Z_i R_i}{\hat{e}(X_i)} - \frac{(1 - Z_i) R_i}{1 - \hat{e}(X_i)} \right\}$$

where $R_i = Y_i - \mu_{Z_i}(X_i)$ denotes the residual from outcome modeling.

Outcome Modeling

- ▶ Outcome modeling approach: Specify a regression model for the (potential) outcome on the treatments and covariates $\mu_z(X)$
- ▶ Fit $\mu_z(X)$ to the observed data and obtain fitted potential outcomes $\hat{\mu}_z(X_i)$ ($z = 0, 1$), i.e. impute the missing potential outcomes
- ▶ All estimands can be estimated based on $\hat{\mu}_z(X_i)$:
 - ▶ For ATE, the outcome-modeling estimator is

$$\hat{\tau}^{\text{ATE}} = \sum_{i=1}^N Z_i(Y_i - \hat{\mu}_0(X_i)) + (1 - Z_i)(\hat{\mu}_1(X_i) - Y_i) / N$$

- ▶ For ATT, the outcome-modeling estimator is

$$\hat{\tau}^{\text{ATT}} = \sum_{i=1}^N Z_i \{Y_i - \hat{\mu}_0(X_i)\} / N_1$$

- ▶ For CATE $\tau(x)$, the outcome-modeling estimator is

$$\hat{\tau}(x) = \hat{\mu}_1(x) - \hat{\mu}_0(x)$$

Outcome Modeling: Linear models

- ▶ A common choice of outcome model is the linear regression:

$$\mu_z(x) = \alpha_z + \beta_z X_i + \epsilon_{z,i}$$

- ▶ For linear models, fitting one model for each z is equivalent to adding an interaction term XZ ; not the case for nonlinear models
- ▶ Then the outcome-modeling estimator of ATE has the exact form as the ANCOVA estimator in randomized experiments

$$\hat{\tau}^{\text{ATE}} = \left\{ \bar{Y}_1 - \hat{\beta}_1 (\bar{X}_1 - \bar{X}) \right\} - \left\{ \bar{Y}_0 - \hat{\beta}_0 (\bar{X}_0 - \bar{X}) \right\}$$

$\hat{\beta}_z$: OLS estimate of the coefficient of Z in the regression $\mu_z(x)$

- ▶ Unlike randomized experiments, the estimator is not consistent if the linear model is misspecified
- ▶ Variance: delta method or bootstrap
- ▶ (Posterior) inference in Bayesian paradigm is automatic (more in later Chapter on Bayesian)

A Toy Example

- ▶ Population: patients with heart attack
- ▶ Treatment Z : 1 surgery; 0 medical
- ▶ Outcome Y : prognosis after 1 month
- ▶ Single covariate X - severity: 0 severe; 1 mild-average. All other covariates are matched between groups
- ▶ Sample: N patients admitted in a hospital
- ▶ Goal: (1) estimate the effect of surgery comparing to medication; (2) predict the prognosis of a new patient
- ▶ It happens to be: in the observed sample, all patients with $X = 0$ get $Z = 1$, and all patients with $X = 1$ get $Z = 0$

A Toy Example

- ▶ An idiosyncratic way is to write down a linear regression model for the observed data:

$$Y \sim a + bZ + cX$$

- ▶ For goal (1): fit the model to the sample, and the OLS coefficient of b is the “treatment effect”
- ▶ For goal (2): for a new patient, plug in Z and X to get a predicted Y
- ▶ **Question:** what if the new patient is with $(Z = 1, X = 1)$ or $(Z = 0, X = 0)$?
- ▶ What is odd here?

A Toy Example

- ▶ There is no interaction $Z * X$ in the model
- ▶ No interaction: effectively, but implicitly, assuming the effect of Z is additive (equivalently homogenous effects)
- ▶ Moreover, there is a complete lack of overlap in X between the two groups in the observed data: $Z * X = 0$ for all units
- ▶ Therefore, even if there is an interaction term, there is no information in the data to estimate the coefficient
- ▶ Regression itself does not take the lack of overlap into account; via extrapolation based on an untestable assumption (homogeneity), the previous model gives a—most likely wrong—point prediction
- ▶ **Take home message:** Regression (or any model) comes with a package, you need to know and acknowledge what assumptions—explicit or implicit—come with that model

Outcome Modeling: Importance of Overlap

- ▶ One can use a wide range of outcome models besides linear models for $\mu_z(X)$, frequentist or Bayesian, machine learning, etc.
- ▶ Key decision on model specification:
 - ▶ Two separate models for each treatment group vs. one unified model with treatment indicator? Equivalence with linear models, but not so for non-linear models
 - ▶ Case dependent, for the latter, crucial to include treatment-covariate interactions
- ▶ If the imbalance of the covariates between the two groups is large, the model-based results heavily relies on extrapolation in the region with little overlap, which is sensitive to the model specification assumption
- ▶ We do not know the exact nature of dependence of the assignment on the covariates, resulting in increased sensitivity to model and *a priori* assumptions

Strategies to Reduce Model Sensitivity

- ▶ To mitigate model dependence, two strategies:
 - (1) design - **balance covariates**, (2) analysis- **flexible models**
- ▶ Best strategy is to use both jointly (double-robust or double learning): first balance covariates in the design stage, then use flexible models in the analysis stage
- ▶ Balance covariates (often involve propensity scores)
 - ▶ Matching
 - ▶ Weighting
- ▶ Flexible models (revisit in the Chapter on HTE)
 - ▶ Semi-parametric models (e.g., power series)
 - ▶ Machine learning methods (e.g., tree-based methods (CART, random forest), boosting)
 - ▶ Bayesian non- and semi-parametric models (e.g., Gaussian Processes, BART, Dirichlet Processes mixtures)

Outcome Modeling Approach: Remarks

- ▶ Pros

- ▶ General: once the potential outcomes are imputed based on the model, any estimand can be estimated
- ▶ Many advanced and flexible models, particularly in complex data, can be used

- ▶ Cons

- ▶ No amount of fancy models can solve the fundamental problem of causal inference: structural missing data (at least half of the data)
 - ▶ Over reliance on outcome model sometimes obscure the crucial role of covariate overlap and balance
- ▶ Ensuring covariate overlap and balance is the key for outcome modeling in causal inference – the key distinction between prediction and causal tasks

References

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